

INTRODUCTION In 1992 Richard Wilkinson published a widely cited paper in the British Medical Journal, arguing that in rich countries higher levels of income inequality lowered average life expectancy.¹ Since then, a growing body of research has explored whether the level of income inequality in a society predicts average health or longevity. A second, separate body of research has explored income disparities in these outcomes. These two traditions should be seen as complementary, because understanding the effects of income inequality on individual health requires us to consider its relationship to both average health and disparities in health. Three promising directions for research are suggested, all of which focus on understanding the mechanisms that link income inequality to average health and health disparities. First, we need more detailed investigation of who is affected by income inequality and how they are affected, since effects are unlikely to be uniform. Second, we need research that takes account of effects that unfold over time, since time lags can be long as well as short. If rising income inequality results in political capture by the rich, for instance, that will affect policy after lags of uncertain but varying duration, and policy changes will in turn affect individuals' health and mortality after lags of uncertain duration. Third, research must distinguish between direct and indirect effects of income inequality. In this report, we describe direct and indirect effects and note prominent theories that predict such effects. We then summarize empirical evidence on the relationships among income inequality, average life expectancy, and socioeconomic disparities in life expectancy.

DIRECT AND INDIRECT EFFECTS OF INCOME INEQUALITY Direct effects of income inequality operate through changes in individuals' own income. Indirect effects operate through changes in other people's income, which change a society's political and economic institutions, as well as its customs and ideals. Such broad social changes can, in turn, alter an individuals' incentives and behavior, even if their own incomes have not changed. Indirect effects can change either the average level of health or the slope of the relationship between individual income and health.

Direct effects. By definition, a rise in income inequality increases purchasing power disparities. If more income improves health, larger income gaps between the rich and poor should result in larger health gaps between rich and poor. In addition, the health of the rich benefits less than the health of the poor from each extra dollar of income. In other words, the line connecting income and health is concave downward. Thus, if all else is equal, transferring money from poorer groups to richer groups will reduce average health. At least two theories relate direct effects of rising income inequality to average health and health disparities. First, rising income inequality at both the individual and the community level is likely to increase inequality in health-producing material resources, such as nutrition, housing conditions, and education.² Second, because material deprivation imposes a cognitive "bandwidth tax" that interferes with decision-making and long-term planning,³ rising income inequality may take a larger cognitive toll on those nearer the bottom of the income distribution. As Figure 1 shows, direct effects depend on whether income inequality grows at the top or the bottom.

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